CREATING LINEAR MODELS FROM WORD PROBLEMS

Digital SAT Style Practice Questions | Difficulty Mix

DOCUMENT 1 — QUESTIONS ONLY | Total Questions: 50

Difficulty Key:	EASY	MEDIUM	HARD	GRID-IN

Q1. EASY

A taxi company charges a flat fee of $3.50 at the start of every ride, plus $2.25 for each mile traveled. Which equation represents the total charge C, in dollars, for a ride of m miles?

A) C = 2.25 + 3.50m
B) C = 3.50 + 2.25m
C) C = 5.75m
D) C = 3.50m + 2.25

Q2. EASY

A candle is 18 inches tall when first lit. It burns down at a constant rate of 0.4 inches per hour. Which equation models the height h, in inches, of the candle after t hours of burning?

A) h = 0.4t + 18
B) h = 18t - 0.4
C) h = 18 - 0.4t
D) h = 0.4 - 18t

Q3. EASY

A parking garage charges $4.00 to enter and an additional $1.50 for each half hour parked. A driver parked for h half-hour intervals. Which of the following represents the total amount, in dollars, the driver paid?

A) 4.00h + 1.50
B) 1.50h
C) 4.00 + 1.50h
D) 5.50h

Q4. GRID-IN

The table below shows two values of a linear function f.

x	f(x)
2	11
6	27

What is the value of f(0)?

GRID-IN: Enter your answer in the box. No answer choices.

Q5. EASY

A gym membership costs $40 to join and $25 per month. A competing gym costs $10 to join and $35 per month. After how many months m would the total cost at both gyms be equal?

A) m = 2
B) m = 3
C) m = 4
D) m = 5

Q6. EASY

A baker starts each morning with 240 bread rolls and sells 15 rolls every hour. Which equation gives the number of rolls R remaining after h hours?

A) R = 15h - 240
B) R = 240 + 15h
C) R = 240 - 15h
D) R = 15 - 240h

Q7. EASY

Maria earns $12.50 per hour working at a bookstore. She also received a one-time signing bonus of $75 when she was hired. Which equation represents her total earnings E, in dollars, after working h hours?

A) E = 75h + 12.50
B) E = 12.50 + 75h
C) E = 12.50h + 75
D) E = 87.50h

Q8. EASY

The temperature in a storage room was 62 degrees F at 8:00 AM and rises at a constant rate of 3 degrees F per hour. What will the temperature be at 1:00 PM?

A) 65 degrees F
B) 68 degrees F
C) 72 degrees F
D) 77 degrees F

Q9. EASY

A school fundraiser sells cookies at $2.50 each and receives a $100 donation from a local business. Which equation represents the total funds F, in dollars, raised after selling c cookies?

A) F = 100c + 2.50
B) F = 2.50c - 100
C) F = 2.50 + 100c
D) F = 2.50c + 100

Q10. GRID-IN

A water tank holds 500 gallons. A pump drains 8 gallons per minute. After how many minutes will the tank have exactly 100 gallons remaining?

GRID-IN: Enter your answer in the box. No answer choices.

Q11. MEDIUM

An online streaming service charges $9.99 per month. The company is offering a deal where new subscribers pay a $14.99 activation fee but only $7.49 per month. After how many full months of service does the total amount paid under the promotional plan first become less than the total amount paid under the standard plan?

A) 5
B) 6
C) 7
D) 8

Q12. MEDIUM

A store sells custom T-shirts. The fixed cost to set up a design is $120, and each shirt costs $8 to produce. The store sells each shirt for $20. Which equation represents the store's profit P, in dollars, after producing and selling n shirts?

A) P = 20n - 120
B) P = 12n - 120
C) P = 12n + 120
D) P = 8n - 120

Q13. MEDIUM

The value of a car depreciates linearly. A car purchased for $28,000 is worth $20,500 after 3 years. What will the car's value be, in dollars, 7 years after it was purchased?

A) $10,500
B) $12,000
C) $13,000
D) $15,500

Q14. MEDIUM

A scientist is filling a tank with a solution. The tank currently contains 12 liters and is being filled at 3.5 liters per minute. A second tank contains 47 liters and is being drained at 2.5 liters per minute. After how many minutes will the two tanks contain the same amount of solution?

A) 5
B) 5.5
C) 6
D) 7

Q15. MEDIUM

An electric vehicle travels 4 miles per kilowatt-hour (kWh) of battery. The car must travel 90 miles and currently has no charge. The driver charges the car at a station providing 2.5 kWh per minute. What is the minimum integer number of minutes the driver must charge to complete the trip?

A) 8
B) 9
C) 10
D) 11

Q16. GRID-IN

The function $f(x) = -3x + b$ passes through the point (4, 7). What is the value of b?

GRID-IN: Enter your answer in the box. No answer choices.

Q17. MEDIUM

A plumber charges $85 for the first hour and $55 for each additional hour. An electrician charges $65 for the first hour and $70 for each additional hour. Let h represent the number of additional hours worked beyond the first hour (h >= 0). For what value of h will the two professionals charge the same total amount?

A) h = 1
B) h = 2
C) h = 3
D) h = 4

Q18. MEDIUM

A train departs from City A toward City B at 75 miles per hour. At the same time, a second train departs from City B toward City A at 60 miles per hour. The two cities are 405 miles apart. After how many hours will the trains pass each other?

A) 2.5
B) 3
C) 3.5
D) 4

Q19. MEDIUM

A factory's total cost C to produce w widgets per day is given by $C = 6w + 450$. The revenue R from selling w widgets is $R = 14w$. What is the minimum number of widgets the factory must sell in one day to make a profit (revenue greater than cost)?

A) 55
B) 56
C) 57
D) 58

Q20. **MEDIUM**

A linear model predicts that a town's population P (in thousands) is given by P = 2.4t + 38, where t is the number of years since 2010. According to this model, in which year will the population first exceed 80,000?

A) 2027
B) 2028
C) 2029
D) 2030

Q21. **GRID-IN**

A cell phone plan charges a monthly base fee of $30 and $0.10 per text message sent. Another plan has no base fee but charges $0.25 per text. A student sends t texts per month. For what value of t will the monthly cost of both plans be equal?

GRID-IN: Enter your answer in the box. No answer choices.

Q22. **MEDIUM**

A construction crew has already built 40 feet of wall and continues at 12 feet per hour. A second crew starts from scratch at 18 feet per hour. After how many hours after the second crew starts will the second crew have built more total wall than the first?

A) 6
B) 7
C) 8
D) 9

Q23. **MEDIUM**

The amount of medication m, in milligrams, remaining in a patient's bloodstream t hours after administration is modeled by m = 500 - 35t. What does the value 500 represent in this context?

A) The rate at which the medication is eliminated per hour
B) The total hours the medication remains effective
C) The initial amount of medication administered
D) The amount remaining after 35 hours

Q24. MEDIUM

Devon currently has \$350 and saves \$45 per week. Sasha currently has \$110 and saves \$80 per week. What is the least integer number of weeks w after which Sasha's total savings will exceed Devon's?

A) 6
B) 7
C) 8
D) 9

Q25. MEDIUM

A swimming pool contains 12,000 gallons. It is being drained at 80 gallons per minute while a hose adds water at 15 gallons per minute. How many minutes will it take to reduce the pool volume to 8,200 gallons?

A) 47.5
B) 56.9
C) 58.5
D) 60

Q26. HARD

A rental company offers two plans for a one-day van rental. Plan A charges \$0.42 per mile with no daily fee. Plan B charges a flat daily fee of \$38.00 plus \$0.18 per mile. At exactly how many miles do the two plans cost the same? For fewer miles than this, which plan is cheaper?

A) 152 miles; Plan A is cheaper for fewer miles
B) 158 miles; Plan B is cheaper for fewer miles
C) 158.33 miles; Plan A is cheaper for fewer miles
D) 158.33 miles; Plan B is cheaper for fewer miles

Q27. HARD

A company's monthly revenue R (in thousands of dollars) is $R = 8.5m + 124$, where m is the month number. A second company's revenue is $R = 12m + 90$. In which month m will the second company's revenue first exceed the first company's?

A) $m = 9$
B) $m = 10$
C) $m = 11$
D) $m = 12$

Q28. HARD

A conference room rents for $75 per hour. Catering costs $18 per person. A company has a budget of $600 for the event and expects 12 attendees. Which of the following correctly gives the maximum number of whole hours h the room can be rented?

A) 75h + 216 <= 600, so h <= 5

B) 75h + 18 <= 600, so h <= 7

C) 75h + 216 <= 600, so h <= 4

D) 93h <= 600, so h <= 6

Q29. HARD

A linear function f satisfies f(2) = 17 and f(9) = 52. What is the value of f(0)?

A) 3

B) 5

C) 7

D) 9

Q30. HARD

The subscribers S to a newsletter t weeks after its launch is S = 120t + 80. A competitor launched at t = 4 (from the first newsletter's perspective) and gains 180 subscribers per week. For what value of t will the competitor's count first equal the first newsletter's count?

A) t = 10

B) t = 12

C) t = 14

D) t = 16

Q31. GRID-IN

A boat travels downstream at (r + 4) mph and upstream at (r - 4) mph. The downstream trip covers 72 miles and the upstream trip covers 40 miles. Both trips take the same time. What is the boat's speed r, in miles per hour, in still water?

GRID-IN: Enter your answer in the box. No answer choices.

Q32. HARD

The cost C to produce x units is C = 14x + 3,200. Each unit sells for $22. Which of the following correctly identifies the break-even quantity and explains what 3,200 represents?

A) 400 units; 3,200 is the fixed overhead cost

B) 229 units; 3,200 is the variable cost per unit

C) 400 units; 3,200 is the revenue at break-even

D) 145 units; 3,200 is the total profit target

Q33. HARD

A function g is linear and satisfies g(3) = -2 and g(7) = -14. What is the x-intercept of the graph of y = g(x)?

A) (-1/2, 0)

B) (1/2, 0)

C) (1, 0)

D) (7/3, 0)

Q34. HARD

Company X has monthly costs C = 500 + 12n and revenue R = 20n - 200. Net income I = R - C. If the company needs net income of at least $1,000, what is the minimum number of units n it must produce and sell?

A) 212

B) 213

C) 214

D) 215

Q35. HARD

A solar panel generates electricity at 0.3 kWh per hour of sunlight. On a day with 9 hours of sunlight, the solar panel generates exactly 60% of the household's daily electricity usage. What is the household's daily electricity usage, in kWh?

A) 3.2

B) 4.5

C) 4.7

D) 5.4

Q36. HARD

Line l passes through the points (-3, 7) and (5, -1). Line k is parallel to l and passes through (0, -4). At what point do lines k and l intersect?

A) They never intersect because they are parallel
B) They intersect at (1, -5)
C) They intersect at (-3, -1)
D) They intersect at (0, 4)

Q37. GRID-IN

A linear function p is defined by p(x) = ax + b, where a and b are constants. It is known that p(-2) = 13 and p(6) = -11. What is the value of a + b?

GRID-IN: Enter your answer in the box. No answer choices.

Q38. HARD

A company's annual profit P (in millions) is linear in x, the years since 2015. In 2017 (x = 2), profit was $3.6 million. In 2020 (x = 5), profit was $7.2 million. According to the model, in what year was profit equal to $0?

A) 2013
B) 2014
C) 2015
D) 2016

Q39. GRID-IN

An athlete's training plan calls for linearly increasing weekly distances. In week 1 she runs 12 miles, and in week 6 she runs 27 miles. In which week g will she first reach 42 miles?

GRID-IN: Enter your answer in the box. No answer choices.

Q40. HARD

A food truck owner's weekly costs are C = 3.60f + 280, where f is the number of food items sold. He sells each item for $7.50 and wants to save exactly $500 per week after all costs. How many items must he sell per week?

A) 197
B) 198
C) 199
D) 200

Q41. HARD

The table below shows values of a linear function h(x):

x | h(x)

-----+------

-4 | 22

0 | 10

4 | -2

Which correctly identifies the rate of change and the meaning of h(0) = 10?

A) Rate = -3; h(0) = 10 is the x-intercept

B) Rate = -3; h(0) = 10 is the y-intercept

C) Rate = 3; h(0) = 10 is the y-intercept

D) Rate = -12; h(0) = 10 is the y-intercept

Q42. HARD

Plan X: $0.38 per mile, no daily fee. Plan Y: $55 per day + $0.15 per mile. For a one-day rental of exactly 230 miles, which plan is cheaper, and by how much?

A) Plan X is cheaper by $2.10

B) Plan Y is cheaper by $2.10

C) Plan X is cheaper by $5.40

D) Plan Y is cheaper by $5.40

Q43. HARD

A city's emergency fund starts with $4,200,000. The city draws $85,000 per month and deposits a $12,500 monthly grant. After how many complete months will the fund first fall below $3,000,000?

A) 16

B) 16

C) 17

D) 18

Q44. HARD

A researcher models skin health score s as a function of weekly UV exposure hours n: $s = -2.5n + 95$. Which is the best interpretation of the slope -2.5?

A) The score increases by 2.5 for each additional UV hour
B) A person with zero UV exposure has a score of -2.5
C) The score decreases by 2.5 for each additional UV hour per week
D) A score of 95 corresponds to 2.5 hours of UV exposure

Q45. GRID-IN

In the equation $T = kd + b$, T is temperature (degrees F) and d is depth in feet below Earth's surface. If $T = 62$ when $d = 150$, and $T = 92$ when $d = 450$, what is the value of k?

GRID-IN: Enter your answer in the box. No answer choices.

Q46. HARD

A warehouse ships out 14 pallets per day and receives 6 new pallets per day. After exactly 18 days, the warehouse has 100 pallets remaining. How many pallets were in the warehouse at the start?

A) 234
B) 244
C) 252
D) 262

Q47. HARD

A freelance designer's fee is $F = 150 + 45h$, where h is hours of revisions. A client receives a quote of $645. The client then requests 2 additional hours of revisions. What will the new total fee be?

A) $690
B) $725
C) $735
D) $745

Q48. HARD

A company's cumulative sales S (in thousands of units) is given by $S = 3.2q - 1.6$, where q is quarters elapsed. Which best interprets -1.6 in this model?

A) The company sold -1,600 units before launching (not meaningful as stated)
B) Sales decrease by 1,600 units each additional quarter
C) At $q = 0$, the model predicts a deficit of 1,600 units -- a limitation of linear extrapolation
D) The company needs 1,600 more units to break even each quarter

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Q49. HARD

A city charges for water linearly. In January, a resident used 4,200 gallons and paid $63.00. In March, the same resident used 7,800 gallons and paid $108.00. Which equation correctly models the bill B as a function of gallons g?

A) $B = 0.0125g + 10.50$
B) $B = 0.025g - 10.50$
C) $B = 0.0125g - 10.50$
D) $B = 0.025g + 10.50$

Q50. HARD

District 1 budget: $B = 1.8y + 22.4$. District 2 budget: $B = 2.6y + 14.8$. Both B values are in millions and y is years since 2005. In which year will District 2's budget first equal or exceed District 1's budget?

A) 2014
B) 2015
C) 2016
D) 2017

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